



Construction Quantity Calculator

Structural Quantity Report

Project: عمارة صادق قبيضة - 123 م³
Client: العميل عبدالله القبيضة
Engineer: المهندس ماجد القبيضة
Date: 16 September 2026
Version: 1.0

Prepared by

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Concrete Volume	Steel Weight	Total Cost	Elements
60.67 m³	600.6 kg	\$7,000.00	27

Total Concrete	Total Steel	Grand Total
60.67 m³	600.6 kg (0.601 tons)	\$7,000.00

Floor Breakdown			
Floor	Concrete (m³)	Steel (tons)	Cost
Floor 1	10.111	0.100	1,000.11
Floor 2	20.222	0.200	2,000.22
Floor 3	30.333	0.300	3,000.33
TOTAL	60.666	0.601	7,000.00

Isolated Footings					
Volume: 0.858 m³ Steel: 60.0 kg Cost: \$145.81 Elements: 3					
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
Footing A 1	0.01×2.01×3.01	1	0.061	10.0	16.05
Footing B 1	0.02×2.02×3.02	2	0.244	20.0	44.40
Footing C 1	0.03×2.03×3.03	3	0.554	30.0	85.36
TOTAL			0.858	60.0	145.81

Strip Footings					
Volume: 37.562 m³ Steel: 60.0 kg Cost: \$3,816.17 Elements: 3					
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
Strip A 2	1.01×2.01×3.01	1	6.111	10.0	621.06
Strip B 2	1.02×2.02×3.02	2	12.445	20.0	1,264.48
Strip C 2	1.03×2.03×3.03	3	19.006	30.0	1,930.63
TOTAL			37.562	60.0	3,816.17

Raft Foundation					
Volume: 74.265 m³ Steel: 60.0 kg Cost: \$7,486.53 Elements: 3					
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
Raft A 3	2.01×2.01×3.01	1	12.161	10.0	1,226.07
Raft B 3	2.02×2.02×3.02	2	24.646	20.0	2,484.56
Raft C 3	2.03×2.03×3.03	3	37.459	30.0	3,775.90
TOTAL			74.265	60.0	7,486.53



Columns				Volume: 110,969 m³ Steel: 60.0 kg Cost: \$11,156.89 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
Column A 4	3.01×2.01×3.01	1	18.211	10.0	1,831.08
Column B 4	3.02×2.02×3.02	2	36.846	20.0	3,704.64
Column C 4	3.03×2.03×3.03	3	55.912	30.0	5,621.17
TOTAL			110,969	60.0	11,156.89

Beams				Volume: 147,672 m³ Steel: 60.0 kg Cost: \$14,827.25 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
Beam A 5	4.01×2.01×3.01	1	24.261	10.0	2,436.09
Beam B 5	4.02×2.02×3.02	2	49.047	20.0	4,924.72
Beam C 5	4.03×2.03×3.03	3	74.364	30.0	7,466.44
TOTAL			147,672	60.0	14,827.25

Solid Slabs				Volume: 184,376 m³ Steel: 60.0 kg Cost: \$18,497.61 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
SolidSlab A 6	5.01×2.01×3.01	1	30.311	10.0	3,041.10
SolidSlab B 6	5.02×2.02×3.02	2	61.248	20.0	6,144.80
SolidSlab C 6	5.03×2.03×3.03	3	92.817	30.0	9,311.71
TOTAL			184,376	60.0	18,497.61

Hollow Block Slabs				Volume: 221,080 m³ Steel: 60.0 kg Cost: \$22,167.97 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
HollowSlab A 7	6.01×2.01×3.01	1	36.361	10.0	3,646.11
HollowSlab B 7	6.02×2.02×3.02	2	73.449	20.0	7,364.88
HollowSlab C 7	6.03×2.03×3.03	3	111.270	30.0	11,156.98
TOTAL			221,080	60.0	22,167.97

Flat Slabs				Volume: 257,783 m³ Steel: 60.0 kg Cost: \$25,838.33 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
FlatSlab A 8	7.01×2.01×3.01	1	42.411	10.0	4,251.12
FlatSlab B 8	7.02×2.02×3.02	2	85.650	20.0	8,584.96
FlatSlab C 8	7.03×2.03×3.03	3	129.722	30.0	13,002.25
TOTAL			257,783	60.0	25,838.33



Waffle Slabs				Volume: 294,487 m³ Steel: 60.0 kg Cost: \$29,508.69 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
WaffleSlab A 9	8.01×2.01×3.01	1	48.461	10.0	4,856.13
WaffleSlab B 9	8.02×2.02×3.02	2	97.850	20.0	9,805.04
WaffleSlab C 9	8.03×2.03×3.03	3	148.175	30.0	14,847.52
TOTAL			294,487	60.0	29,508.69

Bill of Quantities (BOQ)

عمارة صادق قبيظة - 123 م³

#	Type	Floor	Label	L×W×H (m)	Qty	Vol/unit	Total Vol	Steel kg	Cost
1	F	Floor 1	Footing A 1	0.01×2.01×3.01	1	0.061	0.061	10.0	16.05
2	F	Floor 1	Footing B 1	0.02×2.02×3.02	2	0.122	0.244	20.0	44.40
3	F	Floor 1	Footing C 1	0.03×2.03×3.03	3	0.185	0.554	30.0	85.36
4	SF	Floor 2	Strip A 2	1.01×2.01×3.01	1	6.111	6.111	10.0	621.06
5	SF	Floor 2	Strip B 2	1.02×2.02×3.02	2	6.222	12.445	20.0	1,264.48
6	SF	Floor 2	Strip C 2	1.03×2.03×3.03	3	6.335	19.006	30.0	1,930.63
7	RF	Floor 3	Raft A 3	2.01×2.01×3.01	1	12.161	12.161	10.0	1,226.07
8	RF	Floor 3	Raft B 3	2.02×2.02×3.02	2	12.323	24.646	20.0	2,484.56
9	RF	Floor 3	Raft C 3	2.03×2.03×3.03	3	12.486	37.459	30.0	3,775.90
10	C	Floor 1	Column A 4	3.01×2.01×3.01	1	18.211	18.211	10.0	1,831.08
11	C	Floor 1	Column B 4	3.02×2.02×3.02	2	18.423	36.846	20.0	3,704.64
12	C	Floor 1	Column C 4	3.03×2.03×3.03	3	18.637	55.912	30.0	5,621.17
13	B	Floor 2	Beam A 5	4.01×2.01×3.01	1	24.261	24.261	10.0	2,436.09
14	B	Floor 2	Beam B 5	4.02×2.02×3.02	2	24.524	49.047	20.0	4,924.72
15	B	Floor 2	Beam C 5	4.03×2.03×3.03	3	24.788	74.364	30.0	7,466.44
16	SS	Floor 3	SolidSlab A	5.01×2.01×3.01	1	30.311	30.311	10.0	3,041.10
17	SS	Floor 3	SolidSlab B	5.02×2.02×3.02	2	30.624	61.248	20.0	6,144.80
18	SS	Floor 3	SolidSlab C	5.03×2.03×3.03	3	30.939	92.817	30.0	9,311.71
19	HS	Floor 1	HollowSlab A	6.01×2.01×3.01	1	36.361	36.361	10.0	3,646.11
20	HS	Floor 1	HollowSlab B	6.02×2.02×3.02	2	36.724	73.449	20.0	7,364.88
21	HS	Floor 1	HollowSlab C	6.03×2.03×3.03	3	37.090	111.270	30.0	11,156.98
22	FS	Floor 2	FlatSlab A 8	7.01×2.01×3.01	1	42.411	42.411	10.0	4,251.12
23	FS	Floor 2	FlatSlab B 8	7.02×2.02×3.02	2	42.825	85.650	20.0	8,584.96
24	FS	Floor 2	FlatSlab C 8	7.03×2.03×3.03	3	43.241	129.722	30.0	13,002.25
25	WS	Floor 3	WaffleSlab A	8.01×2.01×3.01	1	48.461	48.461	10.0	4,856.13
26	WS	Floor 3	WaffleSlab B	8.02×2.02×3.02	2	48.925	97.850	20.0	9,805.04
27	WS	Floor 3	WaffleSlab C	8.03×2.03×3.03	3	49.392	148.175	30.0	14,847.52
TOTAL						1,329.052	540.0	133,445.25	

Concrete Mix Design — Material Quantities

Mix Ratio 1 : 2 : 4 (Cement : Sand : Gravel) • Compaction factor 1.54

عمارة صادق قبيظة - 123 م³

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Total Concrete Volume60.666 m³

Dry volume required (×1.54): 93.426 m³

Material Quantities			
Material	Volume (m³)	Equivalent Weight	Ratio Part
Cement	13.347	385 bags × 50 kg	1
Sand	26.693	42,709 kg	2
Gravel	53.386	82,748 kg	4
TOTAL DRY VOLUME	93.426	93.426	1 : 2 : 4

Per 1 m³ of Concrete — Reference Quantities		
Material	Quantity	Details
Cement	7 bags	0.2200 m³ • 50 kg/bag
Sand	0.4400 m³	704 kg
Gravel	0.8800 m³	1,364 kg

Total Concrete	Total Steel	Grand Total Cost
60.666 m³	600.6 kg (0.601 tons)	\$7,000.00